

User Manual: Mike - Robot AI Control Assistant

Overview

Mike is a robotic assistant that controls a Dobot Magician robotic arm. You can control the robot using voice commands, hand gestures, or manual controls. Mike can help you draw shapes, move the robot to specific positions, and perform various actions based on your instructions.

Key Features

1. Voice Control

- Control the robot using your voice commands. You can ask Mike to:
 - Draw shapes
 - Move the robot to the home position
 - Activate or deactivate gesture control mode
 - Connect or reconnect the robot
 - Activate manual control mode

2. Gesture Control

- Control the robot using hand gestures:
 - **V gesture:** Activate suction
 - **Gun gesture:** Deactivate suction
 - **Point gesture:** Move the robot (Up, Down, Left, Right)

3. Manual Control

- Use on-screen buttons or keyboard arrows to move the robot:
 - **X+ / X-:** Move robot along the X-axis
 - **Y+ / Y-:** Move robot along the Y-axis
 - **Z+ / Z-:** Move robot along the Z-axis
 - **Suction:** Turn the suction on or off
 - **Home:** Move the robot to its home position

4. Drawing Shapes

- Ask Mike to draw shapes like:
 - Circle
 - Square

- Rectangle
- Triangle
- Line
- Specify dimensions (e.g., radius for a circle, side length for a square, etc.).

5. Object Recognition

- Mike can recognize objects in the camera feed using YOLO (You Only Look Once) model.
- Ask Mike to identify objects, and it will respond with a list of detected objects and their positions.

Instructions:

1. Starting the Program

Upon starting the program, you will see the main window with the chat interface, camera feed, and control buttons.

Connect the Dobot: Click the "Connect Dobot" button to establish a connection with the Dobot robotic arm.

2. Voice Commands

Click the "  Speak" button or say commands aloud.

Common voice commands:

- **"Connect"**: Connects the Dobot.
- **"Disconnect"**: Disconnects the Dobot.
- **"Home"**: Moves the Dobot to its home position.
- **"Manual Control Mode"**: Activates manual control mode for moving the robot using arrow keys or on-screen buttons.
- **"Gesture Mode"**: Activates hand gesture control.
- **"Draw [shape]"**: Asks Mike to draw a shape, e.g., "Draw circle with radius 5".

3. Gesture Control

To activate gesture control, click the "Hand Gesture Mode" button or say **"gesture mode"**.

Common gestures:

- **V gesture**: Turns suction on.
- **Gun gesture**: Turns suction off.
- **Point gesture**: Move the robot in different directions (Up, Down, Left, Right).

To exit gesture mode, click the "  Exit Gesture Mode" button or say "exit gesture mode".

4. Manual Control

Activate manual control mode using the "Manual Control Mode" button or voice command.

Use the on-screen buttons to move the robot:

- **X+ / X-**: Moves robot on the X-axis.
- **Y+ / Y-**: Moves robot on the Y-axis.
- **Z+ / Z-**: Moves robot on the Z-axis.
- **Suction**: Toggles the suction on/off.
- **Home**: Returns the robot to its home position.

You can also use **keyboard shortcuts**:

- **Arrow Keys**: Move the robot in X and Y directions.
- **W/S**: Move the robot along the Z-axis.
- **Spacebar**: Toggle suction.
- **H**: Return to the home position.
- **Escape**: Exit manual control mode.

5. Drawing Shapes

To draw a shape, say something like *****draw a circle with radius 5***** or type it in the input field.

Mike will ask for the dimensions (e.g., side length for squares, radius for circles).

Mike supports the following shapes:

- Circle
- Square
- Rectangle
- Triangle
- Line

6. Object Recognition

To activate object recognition, click the "Object Recognition Mode" button or say "object recognition mode".

Mike will identify objects in the camera feed (e.g., person, cell phone, chair) and tell you what it detects.

You will hear Mike say the object names and their positions relative to you (e.g., "Person in front of you").

Troubleshooting

- Connection Issues: If Mike fails to connect to the Dobot, make sure the robot is properly connected to the computer and retry.
- Voice Command Issues: Ensure the microphone is working and try speaking clearly.
- Gesture Recognition: Ensure the camera is placed properly to detect hand gestures clearly.

Exit/Shutdown:

To exit the program, simply close the window.

If you want to stop any ongoing operation (gesture or manual control), click the appropriate buttons to deactivate the modes.